Task 5: Exploratory Data Analysis (EDA)

Titanic Passenger Dataset — Findings Report

Objective: Extract insights using visual and statistical exploration, as requested in the internship task.

1. Tools and approach

The accompanying Jupyter Notebook uses Pandas for data inspection and summaries, Matplotlib and Seaborn for visualizations, and descriptive statistics to inspect distributions, missing values, relationships, and potential anomalies. The analysis includes info(), describe(), value_counts(), histograms, boxplots, scatterplots, a correlation heatmap, and a pairplot.

2. Data quality observations

The Titanic dataset contains passenger and voyage variables such as survival status, sex, age, passenger class, fare, family aboard, embarkation port, and cabin/deck information. Some fields contain missing values—particularly deck, age, and embarked—so the notebook reports missing counts and percentages before interpreting results. Missing values should be handled according to the question being studied rather than filled indiscriminately.

3. Visual analysis and observations

Analysis / plot	What it helps reveal
Survival count plot	Compares the number of passengers who survived and did not survive.
Age histogram and boxplot	Shows age distribution, spread, and possible extreme values; only recorded ages appear.
Survival rate by sex	Shows substantially higher observed survival among women than men.
Survival rate by passenger class	Shows higher observed survival for first-class passengers than lower-class passengers.
Age vs fare scatterplot	Shows the spread of fares across ages and survival groups; points overlap and high fares may stand out.
Correlation heatmap and pairplot	Summarize relationships among numeric variables; correlations should be interpreted cautiously.

4. Main findings

• Survival outcomes differed across passenger groups; sex and passenger class were strongly associated with survival in the observed data.

• Age and deck/cabin information contain missing values, which affect how much of the dataset can be used for some plots.

• Fare is typically right-skewed, with some unusually high fares. Outliers should be checked rather than removed automatically.

• A correlation heatmap is useful for quickly reviewing pairwise numeric correlations; a pairplot shows distributions and pairwise relationships.

• These patterns are descriptive associations and do not prove that any single variable caused survival.

5. Conclusion

EDA is an important first step because it helps understand the structure and quality of a dataset before modelling. This project inspects summary statistics, missing values, distributions, outliers, and relationships using the tools specified in the task. Exact counts, rates, and plots are calculated in the notebook when it is run against the dataset.

6. Dataset and reproducibility

Dataset: public Seaborn-data Titanic CSV (https://github.com/mwaskom/seaborn-data/blob/master/titanic.csv). The notebook downloads the dataset automatically when internet access is available; alternatively, place a local titanic.csv next to the notebook. See README.md for setup instructions.